INTERNATIONAL BUSINESS PLAN

GRÜNER KARTON

Biodegradable Cardboard Packaging

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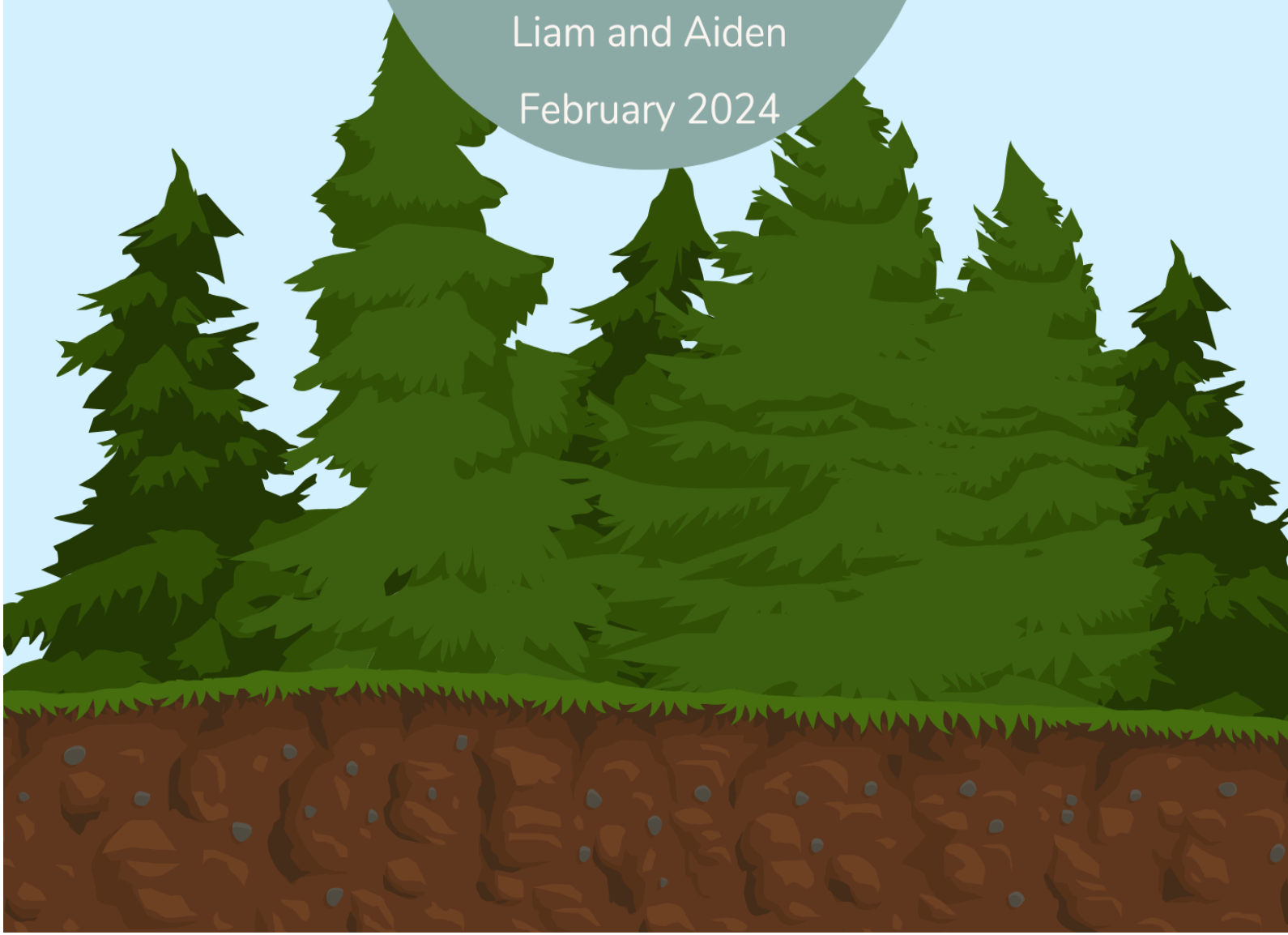


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I. EXECUTIVE SUMMARY

Grüner Karton approaches sustainable packaging through renewable and repurposed cardboard. Our mission addresses growing consumer and industry demand for eco-friendly materials, emphasizing the reduction of environmental impacts such as overconsumption and waste without compromising product quality and functionality.

Our research and development have focused on packaging with renewable and repurposed materials like sawdust and selling it for lower prices thanks to cheap material supply. This unique green product appeals to various industries in Germany. Our transforming process begins by taking leftover sawdust from local wood mills like Binderholz and turning it into pulp with a hydropulper. Then, we add biodegradable PVA glue to create a natural mixture which is then dried and cut with a Fourdrinier machine. This process results in a durable, biodegradable cardboard for packaging our customer's products and for shipping operations.

Our specialized product has a longer lifespan as it can be reused or composted into fertilizer for sustainability. This and its lower selling prices attract businesses to incorporate our sustainable product in their transportation operations. Agricultural businesses, in particular, would greatly benefit from our product, as they can take full advantage of its eco-friendly shipping features and compostable properties. By taking advantage of renewable resources, we provide an solution to traditional packaging and shipping methods that rely on non-recyclable and resource-intensive materials.

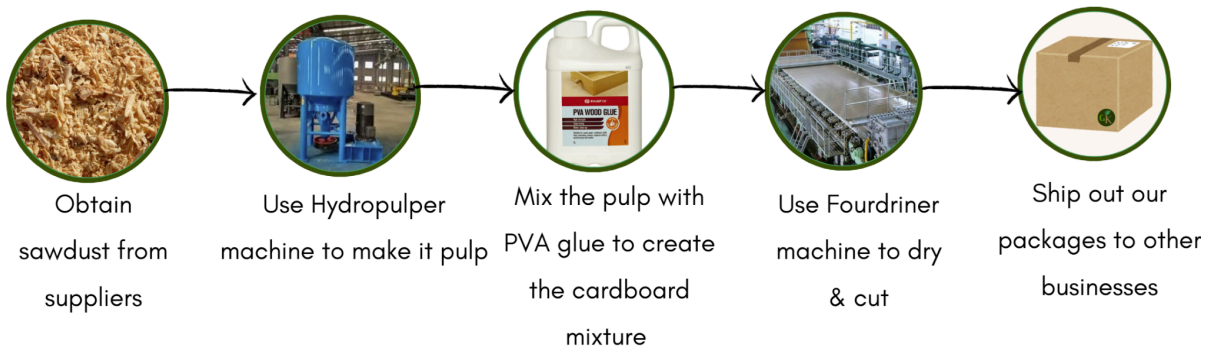


Fig 1: Process of manufacturing

Our product not only commits to sustainability and to customer priority through versatility in customization. Grüner Karton's customization benefits our customers and strengthens our business in several key ways. By offering adjustable dimensions and tailored printing options, we expand our market reach, providing a range of industries and businesses with green packaging needs. This flexibility allows Grüner Karton to be a



customer-focused provider, setting us apart from competitors who may offer more rigid, one-size-fits-all solutions. In addition, our customization services allow us to make additional revenue through premium pricing for its features, such as branded printing or specialized dimensions.

Grüner Karton's workforce includes milling operators, general managers, and truck drivers, each playing a role in maintaining operations. Nine milling operators work in three shifts, ensuring equipment operates at full capacity, distributing workload evenly. The company employs four truck drivers to operate its fleet of trucks in shifts, maximizing distribution. This setup ensures steady production, reliable shipping, and the flexibility to meet high demand, all while maintaining high-quality standards.

Grüner Karton addresses Germany's packaging waste issue, with 18.9 million tonnes generated annually, far above the EU average. By repurposing sawdust from local wood mills, into high-quality cardboard, we eliminate sawdust waste while creating biodegradable, compostable packaging. Using hydropulpers and Fourdrinier machines, our process turns unused byproducts into sustainable materials, aligning with environmental goals. Our solutions also reduce overconsumption of natural resources by eliminating the need for additional timber logging. By using leftover materials, Grüner Karton maximizes material supply and minimises impacts of deforestation, such as biodiversity loss and higher carbon emissions. Additionally, our biodegradable packaging can be composted into fertilizer, offering a sustainable alternative to traditional fertilizer production, which releases harmful byproducts. This feature supports industries like agriculture, enabling businesses to contribute to the economy while reducing their carbon footprint. By addressing waste, resource overconsumption, and fertilizer shortages, Grüner Karton promotes sustainability and long-term environmental resilience.

Grüner Karton's operations demonstrate the environmental and economic advantages of adopting renewable cardboard packaging. Our approach:

- Reduces waste generated during production and typical shipping operations.
- Cuts reliance on raw and fresh materials, conserving natural resources.
- Meets industry standards for recyclability and biodegradability.

The market shows a strong and growing preference for sustainable solutions, particularly evident in Germany. Our choice for this country focuses on Germany's growing e-commerce market which encourages more shipping traffic, ultimately creating a growing

demand for Grüner Karton. More specifically, a potential location for our facility would be Leipzig, a city in eastern Germany known for its economic growth. Leipzig is near forests and wood mills, making access to sawdust supply easy. In addition, this location is near an intersection of major highways, providing access to distribution to cities.

To effectively reach businesses such as grocery stores and logistic companies and ensure efficient delivery, Grüner Karton developed various distribution channels. We will supply local post offices, retail chains, and use our fleet of trucks. Our efficient distribution methods, including compact 2D packaging for increased truck capacity, combined with direct sales and digital marketing initiatives, enhance accessibility while promoting sustainability.

As we continue to develop our methods and scale production to cities with higher shipping traffic such as Munich, our renewable and repurposed cardboard packaging will set a new standard in sustainable solutions. By prioritizing environmental consciousness, we set standards in the green economy, paving towards a sustainable future.

II. ANALYSIS OF THE INTERNATIONAL BUSINESS SITUATION

Trading Company's Economic System

Over the years, Germany has evolved into one of Europe's most industrialized and economically advanced nations. With a GDP per capita of €51,203.55 as of 2023, the country possesses a strong middle class prioritizing sustainable living (Data Commons, n.d.). Nearly one-third of Germany's land is covered in forests, producing an abundance of leftover wood pulp from local mills, which our business can repurpose into eco-friendly packaging. With nearly 11.4 million hectares of forested land, Germany ranks among the top timber producers in Europe (Deutschland.de, 2018). Many of the contributions from the country's GDP is from exports. Germany's primary exports include automobiles, machinery, and medicine, while its major trading partners include USA, France, and Netherlands (TradelmeX, 2024). Germany's economy is also heavily supported by its industrial sector, accounting for 26.84% of its GDP, highlighting the nation's expertise in manufacturing, engineering, and innovation (Global Economy, n.d.). The country's emphasis on renewable energy and sustainability creates opportunities for industries focused on environmental solutions such as our company, Grüner Karton**.** Germany's trade policies and infrastructure, such as its advanced transportation and major shipping ports like Hamburg make it easier for the movement of goods, making it an ideal location for businesses looking to expand and export goods into other countries. The country's strong banking system and low unemployment rates further show its economic stability, ensuring a reliable environment for

investing. In order to thoroughly understand the German demographic and an analysis of the location a SWOT analysis was conducted below.



Trading Company's Government Structure and Stability

According to the Democracy Index 2020, Germany is ranked 5th globally, placing it in the "full democracy" category. This stability provides a solid foundation for businesses, creating an environment where our companies can thrive without fear of political shifts (Democracy Matrix, n.d.).

Germany's political beliefs such as its emphasis on fair labour standards support our company's operations. By ensuring that our business follows ethical labour practices, we align with Germany's values of fair wages, worker protection, and safety. This not only boosts our company's credibility but also improves our reputation with consumers and investors who value ethical business practices.

Laws or Government agencies that affect the Business

Strict environmental rules are enforced by the government, such as the Verpackungsgesetz, a packaging act that encourages recycling and sustainable packaging (Zentrale Stelle Verpackungsregister [ZSVR], n.d.). This law can be an advantage for our company, emphasizing sustainability and reusing products like wood pulp converted into sawdust. According to the Verpackungsgesetz, all businesses that sell packaging must ensure that it is recyclable or composed of recyclable materials. Germany's packaging and trash disposal regulations provide our company both opportunities and challenges. Our business capitalizes on the increasing demand for environmentally friendly products by

producing biodegradable packaging using wood pulp and compostable PVA glue converted to sawdust. Germany's dedication to environmental sustainability improves our brand as an ethical business, which can attract investors and customers who share our values.

In Germany, establishing a business with the local Gewerbeamt (trade office) to receive a corporate licence typically ranges from €10 to €60, depending on the region and the type of business. This country also offers several initiatives aimed at assisting businesses. This price covers the company's basic registration, however, the exact amount depends on several factors, including the location of the registration and the type of business. The fee generally covers the administrative costs associated with the registration process and ensures that the business is legally recognized by the authorities (Merzbach, 2024).

Geographic and Demographic Information

With an estimated population of 83 million, Germany is one of the most populated countries in Europe and is located in central Europe (World Bank, n.d.). Because of its location, the country is one of the core of Europe and international trade. Germany is a perfect place for businesses to expand because of its well-developed infrastructure, including an extensive network of highways, railways, and ports which facilitate domestic and international trade. Although there are big populations in rural areas throughout the nation, most of the population is concentrated in major cities like Berlin, Munich, Hamburg, and Frankfurt.

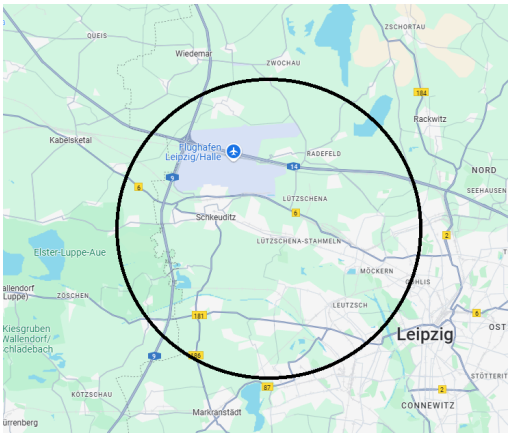
Depending on the industry and region the average income in Germany is approximately €42,000 per year (World Bank, n.d.). The average income is higher in urban places like Berlin and Munich than in rural areas. Nonetheless, the general income level supports a solid middle-class demographic. Businesses like ours that provide higher quality, environmentally cost-effective products are best suited for this demographic while encouraging investors and customers to invest in these products.

Finally, Germany's literacy rate of approximately 99% demonstrates the nation's high level of education and qualified labour force (World Bank, n.d.). The education level shows a more knowledgeable and aware population that can adapt to sustainable practices and the government's rules. As German consumers, this supports our company's goals and emphasis on biodegradable packaging.

Analysis of the Potential Location



A possible location, Leipzig, which is in the eastern part of Germany, is well known for its economic expansion. Leipzig, one of Saxony's closest cities, is located at the intersection of major highways connecting Berlin, Munich, and Prague making it an ideal location for business operations. The forestry in Germany is mostly near the Saxony region, where Leipzig is located. The region's forests produce an estimated 65 million cubic meters of wood annually, making it one of the most productive forestry areas in Europe. Leipzig's location near these forested areas ensures easy access to raw materials such as wood pulp and sawdust, reducing transportation costs. Additionally, many of the sawmills and timber production facilities in Saxony are within a 100 km radius making our operations more efficient (Maier, 2024). Grüner Karton can operate and have facilities at Leipzig's intersection of the A9 and A14 highways, seen on the right. This location is perfect for distribution as it guarantees easy access to major cities and an airport.



Leipzig was selected as the best location for Grüner Karton as it has a mix of accessibility, affordability and suitability for e-commerce operations and transportation. Our location, close to the Leipzig airport which is a hub for cargo carriers like DHL, strengthens our partnership and distribution by prioritizing cargo aircraft, guaranteeing effective logistics and easy access to global shipping routes. To further explore the viability of this location and its market potential, refer to Porter's Five Forces analysis below.

Supplier Power	Buyer Power	Competitive Rivalry	Threat of new entry	Threat of substitute
Leipzig's closeness to the forest ensures a good supply of wood Multiple suppliers reduce dependence on a single supplier	Increased demand for eco-friendly products Key markets include transportation companies, grocery stores, and e-commerce	Few competitors in Leipzig for eco-friendly packaging Lower operational costs due to location give us pricing advantage	Location and partnerships provide market advantages Barriers due to eco-friendly production costs	Businesses may develop low-cost biodegradable packaging that compromises durability or quality

Additionally, without having to pay the far higher operating costs of large cities like Berlin, Munich, or Frankfurt, this guarantees effective access to both local and international distribution networks. Leipzig is also a financially viable alternative due to its lower rent and electricity expenses, which enables us to provide competitive pricing to e-commerce companies looking for affordable packaging options**.** With 5.8% lower overall expenses, Leipzig offers a cost advantage over other larger cities in terms of rent and cost of living (Numbeo, n.d.). This decreased cost benefits not just employees but also companies like ours by lowering the cost of utilities like power and water. These savings, when combined with lower rents and general living expenses, allow us to maintain competitive pricing for our biodegradable packaging while ensuring our operations remain financially sustainable.

III. PROBLEM

i. Packaging Waste

Waste is an active issue that is currently being tackled by all around the world. The need to reduce litter waste has become prominent as the common denominator of almost every industry; packaging, which plays a significant role in the transportation of goods and is a major contributor to this issue. Data from the European Environment Agency's profile for Germany stated they amassed 18.9 million tonnes of packaging waste generated in 2019, 222 kg per capita, far greater than the EU average of 177 kg per capita. This clear difference points out the demand for sustainable packaging that would eliminate waste after use.

ii. Overconsumption of Natural Resources

Overconsumption of natural resources is an issue in Germany due to its large forestry industry. Germany's Federal Statistical Office released data that stated their timber logged remained high at 78.7 million metric cubes in 2022 (Statistisches Bundesamt, 2023). This high consumption of finite natural resources can have larger impacts on climate change, biodiversity loss, and environmental health. Grüner Karton tackles this issue by using repurposed sawdust from wood mills instead of taking new timber for packaging which

would only contribute to this problem. If our recycled materials took over the market, it would make an impact on the total timber logged, effectively lowering natural resource consumption in the country.

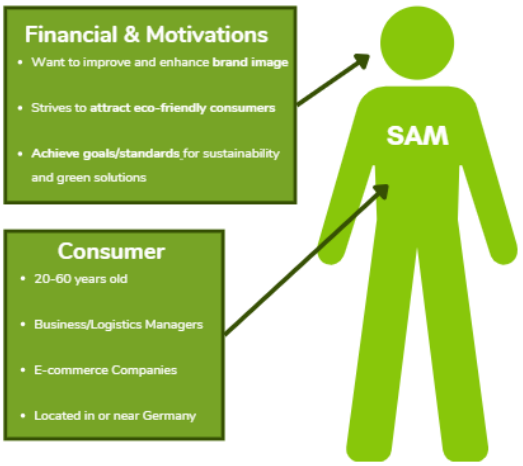
iii. Fertilizer Shortage

The traditional industrial process to create fertilizer called the contact process is used which releases byproducts such as sulfuric acid. These byproducts interact with the atmosphere and induce a higher acidity in the rain. This has a multitude of other negative environmental effects. Grüner Karton uses its unique packaging which allows it to compost into fertilizer due to its wooden origin and biodegradable PVA glue. This prevents the emission of harmful byproducts of the contact process. This transformation creates a more sustainable and eco-friendly alternative that avoids the emission of harmful chemicals in the traditional fertilizer process. This fertilizer would effectively contribute to Germany's strong agriculture industry which consumed 1.3 million tonnes of fertilizer in 2021 (Eurostat, 2023).

IV. CUSTOMER SEGMENTS

By taking advantage of leftover materials from its forestry industry, businesses like ours can transform what was once waste into biodegradable packaging, reducing reliance on plastics. Germany's vast logistics and shipping sectors, supported by its large e-commerce industry create a large opportunity for sustainable packaging. Our target market includes environmentally conscious businesses such as gardening centers, grocery stores, packaging and shipping companies, and governments or NGOs aiming to achieve better sustainability as well as businesses in e-commerce and logistics who require biodegradable packaging solutions.

Take an example, "Sustainable Sam," a 40-year-old operations manager for an e-commerce fulfillment center in Germany. Sustainable Sam is tasked with reducing his company's environmental impact while maintaining the efficiency of logistics operations. His goal is to transition from traditional plastic packaging to biodegradable cardboard solutions that align with his company's sustainability initiatives.



Sam's business makes a substantial contribution to a number of UN Sustainable Development Goals (SDGs) by implementing these solutions. For instance, by encouraging

the use of sustainable materials and decreasing dependency on single-use plastics, the transition to biodegradable packaging directly supports SDG 12: Responsible Consumption and Production. This modification lessens the environmental impact of non-recyclable products and helps reduce waste in landfills. Additionally, by using biodegradable materials that have a lower carbon footprint than standard plastic, the company is lowering greenhouse gas emissions, which connects with SDG 13: Climate Action. For example, using materials made from sawdust and other local forestry leftovers reduces transportation emissions even more, demonstrating the company's commitment to addressing climate change. By taking these significant actions, it appeals to environmentally concerned customers, gives businesses a competitive edge, and further the UN's worldwide goal of building a more sustainable future (UNDP, 2024).

V. UNIQUE VALUE PROPOSITION

Grüner Karton's mission is to create a more sustainable industrial environment and foster eco-friendly product transportation following our SUSTAIN approach (Sustainable, Unique, Sawdust based, Trustworthy, Adaptable, Innovative, Necessary). Our simple but renewable cardboard material made from repurposed sawdust achieves our mission and sets the standard for our environmental values.. This biodegradable packaging meets Germany's sustainability policies and attracts companies prioritizing eco-conscious practices. For instance, agricultural businesses transporting fruits and vegetables can reuse or compost our durable boxes to support their operations. Grüner Karton transforms sawdust into biodegradable, customizable cardboard packaging, offering businesses a practical solution that reduces waste and aligns with modern environmental goals.

VI. SOLUTION

i. Eliminating Sawdust Waste

Our packaging begins with the repurposing of sawdust sourced from local wood mills like Binderholz, Germany's largest timber producer. By turning unused byproducts into high-quality cardboard, we reduce waste of sawdust. Using a hydropulper, the sawdust is processed into pulp and combined with biodegradable PVA glue to create a natural, compostable mixture. Through drying and cutting with a Fourdrinier machine, we produce biodegradable packaging that aligns with modern sustainability goals. This process eliminates any potential waste of sawdust from wood mills as we repurpose this resource.

ii. Reducing Overconsumption of Natural Resources

By utilizing sawdust, Grüner Karton eliminates the need to cut down additional trees, reducing the demand on forestry resources. If our recycled materials were to dominate the

packaging market, it would lead to a meaningful reduction in timber logging for cardboard. This approach conserves natural resources and aligns with Germany's growing demand for eco-friendly practices, providing a sustainable solution to traditional packaging methods. In addition, this shift in the consumption of resources helps minimize the environmental impacts of deforestation, such as biodiversity loss and increased carbon emissions. By using repurposed materials, Grüner Karton promotes long-term environmental resilience and resource efficiency.

iii. Supplying for Fertilizer Shortage

Our cardboard is biodegradable and can be composted into fertilizer, contributing to a circular economy. This feature particularly appeals to industries such as agriculture, which uses compostable materials to create fertilizer, reinforcing sustainable practices. The eco-friendly lifecycle of our product resonates with environmentally-conscious consumers, enabling businesses to showcase their commitment to reducing their carbon footprint. Our product can be resold for fertilizer by businesses for money back instead of being recycled or thrown away, contributing to the agricultural industry and supporting farmers.

VII. CHANNELS

To effectively reach out to clients and customers and while ensuring a smooth delivery, we have developed multiple pathways that prioritize efficiency in distribution, better accessibility for customers, and marketing initiatives.

For distribution, we plan to partner with major distributors such as local post offices, Staples, and other retail chains in Germany. Additionally, by shipping our products with our delivery truck, our business can take advantage of Germany’s small country and interconnected transportation routes. To maximize efficiency and space our packages will be condensed into 2D form, increasing the volume of goods that can be transported into a single truck. These distributors will resell our products to businesses, grocery stores, e-commerce facilities that need reliable packaging. Additionally, partnerships with transportation companies who specialise in grocery and bulk shipping will enable us to a wider market while offering discounts to encourage bulk purchases of our product.

Direct sales to specific businesses will also expand our customer base. Companies that require customized packaging such as grocery stores will benefit and be performed by our Zebra ZT231 printing machine, which has the ability to tailor designs, logos or labels to suit the branding needs. Furthermore, in order to market our products within our company, we will use digital channels, including an online platform to provide ordering and secure payment options.

Furthermore, for our marketing initiatives, we will host specific showcases and product samples at highly populated public areas and industry specific venues or meetups from time to time. These meetups would help our company gain a better reputation, word of mouth, and better foot traffic. The specific trade shows can give our clients and other businesses the history of our business and how our products were made, and sampling our products will lead to more customers. Our digital marketing strategy can consist of search engine optimization where we can have high ranking keywords like “biodegradable packaging solutions” to improve the visibility of our business. Social media platforms are used to engage with potential clients while highlighting the environmental benefits of our products through posts, ads and collaborations. Additionally, distributing flyers and sponsoring events in high foot traffic areas will increase our brand awareness. Rewards programs, rebates and loyalty discounts will encourage other businesses to repeat purchases and build a relationship with our clients.



VIII. REVENUE STREAMS

Our revenue comes from the purchases of our product and additional customization options. Grüner Karton’s revenue model is rooted in the sale of sustainable, durable shipping boxes tailored to the needs of large retail businesses, such as supermarkets. The company generates income through both standard and customized packaging solutions which include dimensions, thickness, material and printing. Grüner Karton focuses on long-term, recurring contracts with businesses, ensuring a predictable revenue stream.

For accurate startup projections of Grüner Karton, we estimated to supply three large retail businesses such as supermarkets. We sell each box without customization at $0.44 CAD (€0.30) and assuming each business uses 250,000 shipping boxes with no customization per month, we would make $113023.14 CAD (€76,095) per month to supply one business. This would total to roughly $338532.06 CAD (€227,923.21) per month in revenue from four large retail businesses.

IX. COST STRUCTURE

Customer Acquisition Costs

Grüner Karton can partner with shipping companies where we supply boxes in exchange for advertising on transportation vehicles. The DHL would be a potential partnership, fitting our needs to advertise in Germany. We will allocate 5% of our cardboard box stock for their logistics operations, in exchange for advertising and promotion on their delivery trucks. This partnership increases brand visibility through exposure on DHL's trucks,

reaching a wide audience across Germany's car traffic. This cost-effective strategy targets businesses in need of packaging and builds credibility by associating with a trusted logistics company. The partnership also aligns with both companies' focus on sustainability, attracting customers who prioritize eco-friendly solutions. If we attract three large retail businesses in our first year, and make our supply and labour costs roughly $8000, so our CAC would roughly be $2666.67 per customer.

Operation Costs

Our four delivery trucks will transport our product directly to our clients as frequently as possible. Due to our folding cardboard boxes, we are able to transport a larger number of stock in one trip. Each truck costs approximately $40,000, depending on the model and its condition, which represents a significant initial investment. In addition, the monthly fuel costs for each truck are around $9,000, assuming each truck's 250L tank needs two refills per month. These fuel costs can fluctuate based on fuel prices and route efficiency. Beyond the fuel costs, there are additional maintenance and insurance expenses, which can vary depending on the truck's age and usage. Efficient route planning and regular maintenance will help lower these expenses, ensuring cost-effectiveness in our distribution.



Human Resource Costs

The human resource costs for Grüner Karton will include salaries for employees in various roles, including milling machine operators, general managers, and truck drivers.The milling operators at Grüner Karton play a vital role in ensuring the smooth functioning of production, with a total of 9 milling operators working in 3 shifts of 8 hours per day to maximize productivity. SalaryExport states in Germany, each operator is paid $27.26 CAD per hour (€18.14), and there are always at least 3 operators at any time working 355 days a year, excluding holidays. Therefore, the total salary for all the operators totals to roughly $696,765.60 CAD. Three operators at all times ensures that the equipment operates at full capacity while allowing operators to maintain machinery without overburdening a single individual. With this setup, production remains steady, downtime is minimized, and any technical issues can be promptly addressed, ensuring high-quality output and maximizing efficiency.

For management, Grüner Karton employs two general managers, each with an annual salary of $90,000 CAD, bringing the total annual salary expense for the management team to $180,000 CAD. Two general managers would be optimal for distributing and balancing the responsibility of the 24-hour operations.

Four truck drivers are critical to the distribution process, earning €50,000 per year ($75,000 CAD per year) according to the Economic Research Institute. With a fleet of four trucks, the company will need to hire enough drivers to ensure consistent and efficient deliveries. The total annual salary for four truck drivers, assuming one driver per truck, will be $300,000 CAD. Four truck drivers are essential for maintaining an efficient distribution process, as each driver can focus on operating one truck, ensuring continuous deliveries in shifts. This arrangement allows the company to maximize the utilization of its trucks, minimizing downtime and optimizing shipping times. With four drivers covering various shifts, the company can handle and become flexible during high-demand periods for customer satisfaction.

The overall human resource costs will therefore include salaries for the production staff, general managers, and truck drivers, adding up to significant expenses on a monthly and annual basis. Proper allocation of labour costs, ensuring shifts are optimized for productivity, and managing staffing levels will be essential for maintaining efficient operations while controlling expenses. Therefore,the total cost for human resources will be $1,176,765.60 excluding any bonuses or additional money.



Fig 2: Fourdrinier Machine (Exporters India, n.d)



Fig 3: Hydropulper Machine (CNBM International, n.d.)

Additional Costs

The operational expenses supporting this business are essential for its success. Rent for the 1200 square meter facility, leased at a rate of €10.50 ($15.59 CAD) per square meter, totals $18,884.31 CAD per month. Utility expenses include $1,049.06 CAD monthly for electricity consumption of 39 kWh and $149.84 CAD monthly for 50,000 litres of water. Additionally, office furniture and supplies, including stationery, are prioritized to ensure a well-equipped and efficient workspace which will amount to around $200. These combined costs reflect the financial commitment necessary to achieve operational excellence while advancing our goals in sustainable packaging.

X. DETAILED FINANCIALS

A. Projected income and expenses

Projected income statements by month for the first year's operation

The income statement below outlines how Grüner Karton is projected to generate revenue during its first year of operating. With an annual revenue of $2,217,807.72 CAD and monthly growth in revenue shows financial stability. From January to August it is shown to have steady increases, however the significant increase in revenue from August to October can be explained by new partnerships and expanding our market. As Grüner Karton gains customers and creates a better reputation, we expect to secure more partnerships from large retail and grocery stores. Next, the net income is projected to be $383,150.76 CAD, which serves as a strong foundation towards paying off our loan and enhancing our facility and equipment.

Grüner Karton	Annual	January	February	March	April	May	June	July	August	September	October	November	December
Revenue	$2,217,807.72	$30,563.22	$73,665.69	$98,589.85	$114,589.85	$118,723.45	$124,443.73	$143,694.20	$147,555.19	$242,463.54	$371,420.00	$375,130.00	$376,969.00
Cost of Goods Sold	$206,956.44	$2,854.17	$6,873.01	$9,198.43	$10,691.23	$11,076.90	$11,610.60	$13,406.67	$13,766.90	$22,621.85	$34,653.49	$34,999.63	$35,203.56
Gross Profit	$2,010,851.28	$27,709.05	$66,792.68	$89,391.42	$103,898.62	$107,646.55	$112,833.13	$130,287.53	$133,788.29	$219,841.69	$336,766.51	$340,130.37	$341,765.44
Operating Expenses													
Rent	$226,611.72	$18,884.31	$18,884.31	$18,884.31	$18,884.31	$18,884.31	$18,884.31	$18,884.31	$18,884.31	$18,884.31	$18,884.31	$18,884.31	$18,884.31
Utilities	$189,673.20	$7,042.00	$12,165.07	$13,069.69	$14,065.07	$16,075.83	$16,269.83	$17,102.55	$17,620.42	$18,865.29	$19,099.33	$19,110.43	$19,187.69
Salaries	$1,176,765.60	$98,063.80	$98,063.80	$98,063.80	$98,063.80	$98,063.80	$98,063.80	$98,063.80	$98,063.80	$98,063.80	$98,063.80	$98,063.80	$98,063.80
Interest Expense	$25,650.00	$2,137.50	$2,137.50	$2,137.50	$2,137.50	$2,137.50	$2,137.50	$2,137.50	$2,137.50	$2,137.50	$2,137.50	$2,137.50	$2,137.50
Insurance	$9,000.00	$750.00	$750.00	$750.00	$750.00	$750.00	$750.00	$750.00	$750.00	$750.00	$750.00	$750.00	$750.00
Total Operating Expenses	$1,627,700.52	$126,877.61	$132,000.68	$132,905.30	$133,900.68	$135,911.44	$136,105.44	$136,938.16	$137,456.03	$138,700.90	$138,934.94	$138,946.04	$139,023.30
Net Income	$383,150.76	-$99,168.56	-$65,208.00	-$43,513.88	-$30,002.06	-$28,264.89	-$23,272.31	-$6,650.63	-$3,667.74	$81,140.79	$197,831.57	$201,184.33	$202,742.14

Projected Balance Sheet - End of First Year

The chart below describes our projected balance sheet for the end of the first year. With total assets being $612,485.76 CAD, large cash reserves of $395,150.76 CAD, and equipment valued at $174,500.00 CAD, the business is well–positioned for operations. Our liabilities were projected at $502,835.00 CAD, rooting from a $500,000 CAD bank loan and interest, leaving owner's equity at $108,650.76 CAD in the first year. This shows Grüner Karton's ability to balance its assets and liabilities effectively while maintaining equity.

Grüner Karton			
Assets		***Liabilities***	
Cash	$395,150.76	Accounts Payable	$3,835.00
Inventory	$2,835.00	Bank Loan	$500,000.00
Supplies	$40,000.00		
Equipment	$174,500.00		
Total Assets	$612,485.76	Total Liabilities	$503,835.00
Owner's Equity		$108,650.76	

Projected Three-Year Plan

The projected three year plan shown below represents the growth of Grüner Karton growth in both revenue and profitability, demonstrating the potential of the business. Operating costs see a steady increase which grows exponentially around $100,000 CAD per year. This shows Grüner Karton's focus on controlling our expenses even as we expand our operations. These cost expenses allow us to improve our revenue which translates into net income. Furthermore, as for the revenue it also sees an exponential increase due to the expansion of its customer base, and entering retail markets. As the business gains more traction the growth showcases the higher demand for sustainable packaging solutions.

Grüner Karton	Year 1	Year 2	Year 3
Revenue	$2,217,807.72	$4,259,703	$7,012,171
Total Cost of Sales	206,956.44	$393,186.39	$502,452.75
Gross profit	$2,010,851.28	$3,866,516.78	$6,509,717.90
Gross profit Margin	90.67%	90.77%	92.83%
Total operating cost	$1,627,700.52	$1,701,016.83	$1,812,087.93
Net Income	$383,150.76	$683,537.73	$900,690.42

Planned Growth of Business

By building a production facility that serves companies looking for eco-friendly and biodegradable solutions, Grüner Karton will bring change with sustainable packaging in Leipzig. Grüner Karton will provide individualized, reasonably priced, and environmentally friendly products to both small and major distributors. With a predicted improvement in gross profit margins from 90% in year one to 93% in year three, we hope to raise gross sales by 115% by the second year. The development of customized models that encourage large purchases and client loyalty is correlated to this increase. These actions will improve cash flow and inventory management when combined with a focus on operational efficiency. The business will use efficient production techniques and make investments in machinery to maximize operational effectiveness in order to handle market fluctuations within Germany.

B. Proposed plan to meet capital needs

Grüner Karton will take out a loan from local banks to fund the initial operations and equipment, which we will then use to expand into other nations.

With revenue of more than $2.2 million in the first year, the company expects a steady improvement in net profitability from an increase of approximately $300,000 CAD in the first year to over $600,000 CAD by the second. A $500,000 bank loan will provide additional funding for upfront expenses such as marketing initiatives, equipment, and

salaries. One of Grüner Karton's debts is this loan that is secured at a 5.13% yearly interest rate by the German government (5-year payback). This corresponds to a manageable cost for funding operations and expansion, with an interest expense of about $2,137.50 per month. The company will also have to pay liability and property insurance each month, which is projected to be around $750. This insurance will guarantee financial protection against any dangers by offering crucial coverage to protect the business's operations and assets.



Grüner Karton will use grants and incentives from organizations such as the German Federal Environmental Foundation, which provides funding for innovative and environmentally sustainable projects like ours. These grants will help finance research, development, and implementation of eco-friendly businesses (German Federal Environmental Foundation, 2021). Through these partnerships, Grüner Karton will be able to access important outside resources and also have a position as a market leader in sustainable packaging.

Beginning in the second year, the $500,000 bank loan will be repaid with $2,137.5 monthly payments. As profitability improves in our business, payments will rise to $5,000 by year three. This ensures that commitments are fulfilled without compromising operational effectiveness by matching anticipated cash flows. The consistent growth in net income and gross profit margin provides investors with a competitive return on their investment. Grüner Karton's financial plan ensures debt repayment and sustained profitability.

XI. KEY METRICS

As a forestry and packaging company that generates revenue from other industries, it is critical that we use the indicators and metrics listed below to assess our ability to increase customer satisfaction, marketing effectiveness, operational efficiency and attraction of new customers.

Customer Retention Rate:

Customer retention rate is calculated and based on the businesses and clients who purchase our products. We can use this as a critical indicator to determine the percentage of businesses that reorder our packaging products over a given time period, as well as to demonstrate their satisfaction and growth of our company. Customer retention rate is calculated as [(# of customers at the end of a period - Customers added within the given period)/ # customers at the start of a period] x 100 (Wren, 2024).

Customer Loyalty:

To assess customer loyalty, our company will use statistics that incorporate aspects of the Net Promoter Score (NPS) and the monthly customer turnover rate to track client loyalty. Through survey replies, this statistic will assess client loyalty while tracking the monthly percentage of lost consumers. The churn rate, which is determined by dividing the total number of customers at the beginning of the month by the number of customers lost at the end of the month, by 100, helps areas that require improvement in retention. Our goal is to keep the turnover rate between 5% and 10% by providing outstanding customer service, implementing reward programs, and using surveys to get useful input. In addition, surveys will determine the likelihood that consumers will suggest our items. Promoters and detractors will be identified from the responses, and the findings will inform churn reduction. Combined, these strategies enable us to improve client happiness and establish enduring customer relations. (Wren, 2024)

Inventory Turnover:

Inventory turnover is a critical metric reflecting how efficiently our company will manage its inventory throughout its operating cycles. It measures the number of times inventory is sold, used, or replaced over a specific period, typically a year or quarter. A high inventory turnover rate shows a business's ability to sell products quickly, therefore reducing holding costs with storage and minimizing the risk of obsolescence. In contrast, a low turnover rate may show overstocking issues or challenges in sales, which could point to less efficient inventory management practices or decreased consumer demand. This metric will help us determine the supply and demand of our current market and plan accordingly to customer preferences and economic changes.

XII. COMPETITIVE ADVANTAGE

Grüner Karton stands out in the packaging industry by using biodegradable materials made from renewable resources such as sawdust. We provide a sustainable method by using leftover sawdust from nearby sawmills to develop eco-friendly packaging solutions, as opposed to competitors who use hybrid products incorporating plastic components

Our manufacturing process uses specialized equipment, such as hydropulpers and Fourdrinier machines, posing a barrier for competitors to enter the market with a similar product. Furthermore, our ability to design fully biodegradable packaging without using hybrid materials such as plastic assures that our product passes environmental standards, something that not all businesses can achieve.

To reach a wider audience, we provide customization options that allow companies to add branding while modifying the dimensions of the packaging without compromising

product's environmental friendliness. This personalization combined with competitive pricing and quality, ensures that our product is tailored to meet customer specific demands. By ensuring quality through customization, we can develop strong relations with clients and strengthen our brand while supporting our sustainability objectives and promoting long-term success.

XIII. CONCLUSION

To finance the launch of our business, Grüner Karton, into Leipzig, Germany, we are requesting an initial investment of $500,000 CAD. Along with operational costs like rent, utilities, and employee pay, this investment will cover important launch costs including $174,500 CAD for equipment and $40,000 CAD for supplies. Our expected monthly running expenses come to $134,877.61, and our estimated annual expenses come to $1,622,332.47. With this money, our business can obtain the resources it needs, establish itself during our first year, and lay the foundation for long-term success. With this support, we plan to establish a strong presence in Leipzig, using the city's closeness to key roadways and forest resources.



Leipzig is the perfect foundation for Grüner Karton because of its location and reasonable cost. Leipzig has far cheaper rent than big cities like Berlin, which lowers operating costs and increases profit margins. Leipzig's annual rent for a facility like ours is about $226,611.72, a significant financial benefit. Additionally, Leipzig's close proximity to major roads enables us to easily service clients throughout Germany and nearby European markets. Our supply chain is further strengthened by the city's easy access to forestry resources, which lowers the cost of transporting raw materials.

Offering reasonably priced, biodegradable packaging that satisfies Germany's environmental standards and contributes to the global movement toward sustainability, Grüner Karton is in a unique position to thrive. We are certain that Grüner Karton will prosper in this industry thanks to our solid financial outlook, location, and environmentally friendly innovation. In addition to receiving large financial rewards, those who invest in Grüner Karton will also help create a more environmentally friendly future. More than just a company, Grüner Karton aims to provide sustainable solutions and dedication to creating a better future for our society and customers.

XIV. BIBLIOGRAPHY

Cost of living comparison between Berlin, Germany and Leipzig, Germany. (n.d.). https://www.numbeo.com/cost-of-living/compare_cities.jsp?country1=Germany&country2=Germany&city1=Berlin&city2=Leipzig&tracking=getDispatchComparison

Customer retention rate + formula: A guide for 2024. (2024, March 27). Zendesk. https://www.zendesk.com/blog/calculate-customer-retention-rate/

Eurostat. (2023, June 30). Consumption of fertilisers in the EU down 2% in 2021. Eurostat. https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20230630-1#

European Environment Agency. (n.d.) Packaging waste generation and treatment in Germany. https://www.eea.europa.eu/publications/many-eu-member-states/germany#:~:text=Packaging%20waste%20generation%20and%20treatment,average%20of%20177%20kg%2Fcap.

Forestry and wood. (n.d.). Federal Statistical Office. https://www.destatis.de/EN/Themes/Economic-Sectors-Enterprises/Agriculture-Forestry-Fisheries/Forestry-Wood/_node.html

German federal Environmental Foundation. Terra Viva Grants. (2021, July). https://www.terravivagrants.org/group-5-cross-cutting/german-federal-environmental-foundation/#

Germany - Place Explorer - Data Commons. (n.d.). https://datacommons.org/place/country/DEU?utm_medium=explore&mprop=amount&popt=EconomicActivity&cpv=activitySource,GrossDomesticProduction&hl=en

Germany Share of industry - data, chart | TheGlobalEconomy.com. (n.d.). TheGlobalEconomy.com. https://www.theglobaleconomy.com/Germany/Share_of_industry/

How large are Germany's forests? Facts and figures. (n.d.). https://www.deutschland.de/en/topic/environment/how-large-are-germanys-forests-facts-and-figures

IBISWorld, Inc. (n.d.). *IBISWorld - industry market research, reports, and statistics*. Copyright © 1999-2024 IBISWorld, Inc. https://www.ibisworld.com/germany/industry/forestry-logging/1475

Low-consistency pulper. Paper Pulp Mill. (n.d.). https://paperpulpingmachine.com/paper-pulper-machine/low-consistency-pulper/

Packaging Act. (n.d.). https://www.verpackungsregister.org/en/foundation-authority/packaging-act

Ranking. (n.d.). https://wwwPackaging Act.democracymatrix.com/ranking

Paper making and the Fourdrinier Machine at best price in Ghaziabad - ID: 3958278. at Best Price in Ghaziabad - ID: 3958278. (n.d.). https://www.exportersindia.com/product-detail/paper-making-and-the-fourdrinier-machine-3958278.htm

Sophia. (2024, November 13). *How to apply for a Gewerbeschein (trade license)*. Accountable. https://www.accountable.de/en/blog/trade-license-germany/

Sustainable development goals. UNDP. (2024). https://www.undp.org/sustainable-development-goals

TradeImeX. (2024, February 28). What are the top 10 exports of Germany? *Infinite*. https://www.tradeimex.in/blogs/top-10-exports-of-germany

World Bank Open Data. (n.d.). World Bank Open Data. https://data.worldbank.org/country/germany

Zebra ZT231 Direct Thermal/Thermal Transfer Printer H-10807 - Uline. (n.d.). https://www.uline.ca/Product/Detail/H-10807/Zebra-Industrial-Barcode-Printers/Zebra-ZT231-Direct-Thermal-Thermal-Transfer-Printer